

NeuralChat

LLM-Powered Chatbot with Login Authentication & Persistent Chat History

PROJECT OVERVIEW

NeuralChat is a full-stack AI chatbot application that integrates **Streamlit** for the web interface, **MongoDB** for persistent storage, and **Ollama (Mistral 7B)** as the local LLM backend. Users register and log in securely; every conversation is stored and restored on the next login — enabling continuous, context-rich dialogues.

KEY FEATURES

- Secure login & registration (SHA-256 hashed passwords)
- Mistral 7B inference via Ollama REST API (localhost:11434)
- Full conversation history persisted in MongoDB with UTC timestamps
- Multi-turn context window: last 20 messages sent per request
- Per-user message stats in dark-mode sidebar
- Zero cloud dependency — runs entirely on local hardware

ARCHITECTURE

Three-tier design: Streamlit frontend → MongoDB auth/storage layer → Ollama LLM. Auth verifies SHA-256 hashes; messages are saved per-turn under the user's namespace; history is hydrated on login and the last 20 turns are bundled with each prompt for coherent dialogue.

GROUP MEMBERS

Name	Student ID	Role
Alice Rahman	2021-CS-01	Lead Dev
Bashir Hossain	2021-CS-02	DB & Auth
Chitra Akter	2021-CS-03	UI / UX
Devraj Mondal	2021-CS-04	Docs & QA

GITHUB REPOSITORY

<https://github.com/your-org/neuralchat>

COURSE & DATE

CSE-4112 · Distributed Systems Lab

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SUPERVISOR

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TECHNOLOGY STACK

Component	Technology	Purpose
Frontend	Streamlit ≥ 1.35	Interactive web UI
LLM Backend	Ollama + Mistral 7B	Local inference REST API
Database	MongoDB ≥ 6.0	User auth + message storage
Auth Layer	SHA-256 + PyMongo	Hashed passwords, unique index
HTTP Client	requests	Calls /api/chat on port 11434

QUICK START

```
# 1. Install Python deps pip install -r requirements.txt # 2. Start MongoDB mongod --dbpath /data/db # 3.
Start ollama & pull model (~4 GB) ollama serve && ollama pull mistral # 4. Launch the app streamlit run
app.py # → Open http://localhost:8501
```